DISABLING AUTO-CREATION FOR AN INDEX FEATURE

As mentioned earlier, Elasticsearch lets us block the automatic creation of indexes by setting the `action.auto_create_index` attribute to `false` (the default is `true`). We can tweak this setting to modify the flag's value by invoking a cluster-wide property change using the cluster settings API. The following listing disables the feature.

Listing 6.2 Disabling automatic creation of indexes

```
PUT _cluster/settings
{
  "persistent": {
    "action.auto_create_index":false
  }
}
```

Updates the settings across the whole cluster

The changes can be persistent or transient.

Shuts down auto-creation

The `persistent` property indicates that the settings will be permanent. On the other hand, using the `transient` property saves the settings only until the next reboot of the Elasticsearch server.

While disabling this feature sounds cool, in reality, doing so is not advisable. We are restricting automatic creation on *any* index, but an application or Kibana may need to create an index for administrative purposes; Kibana often creates hidden indexes. (A dot placed in front of an index name is treated as a hidden index: `.user_profiles`, `.admin` etc.)

Rather than switching off the automatic index creation altogether, there is a way to tweak this property beyond binary options. We can instead provide a selective set of indexes via comma-separated regular expressions that allow (or disallow) this change. Here's an example:

```
action.auto_create_index: [".admin*, cars*, "*books*", "-x*","-y*","+z*"]
```

This setting allows the automatic creation of hidden indexes with `admin` as the prefix, as well as any index prefixed with `cars` or `books` and those following the plus sign (`+`). However, this setting does not allow any index starting with `x` or `y` to be automatically created because the minus sign (`-`) indicates that automatic index creation is disallowed. It also means any other index that doesn't match this pattern will not be instantiated automatically. For example, if we try to index a document into the `flights` index, index creation will fail because the index name doesn't match the regular expressions we just defined (more on this in a bit). Here's the exception issued by the engine:

```
no such index [flights] and [action.auto_create_index]
([.admin*, cars*, *books*, -x*,-y*,+z*]) doesn't match
```

Allowing the server to create the index fosters a rapid development process. However, we rarely go into production without tweaking some of these properties. For example, we may decide to have a strategy of 10 primary shards with 2 replicas per shard, in which case we must change the settings (it would be disastrous to design a search service with just one primary shard). Also, as we learned in chapter 4, Elasticsearch may